Exp No: 27

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BINARY SEMAPHORE IN SYSTEMC

AIM:

To implement and simulate a Binary Semaphore using SystemC for task synchronization.

SOFTWARE REQUIRED:

• Ubuntu (or WSL Ubuntu)

• g++

• CMake

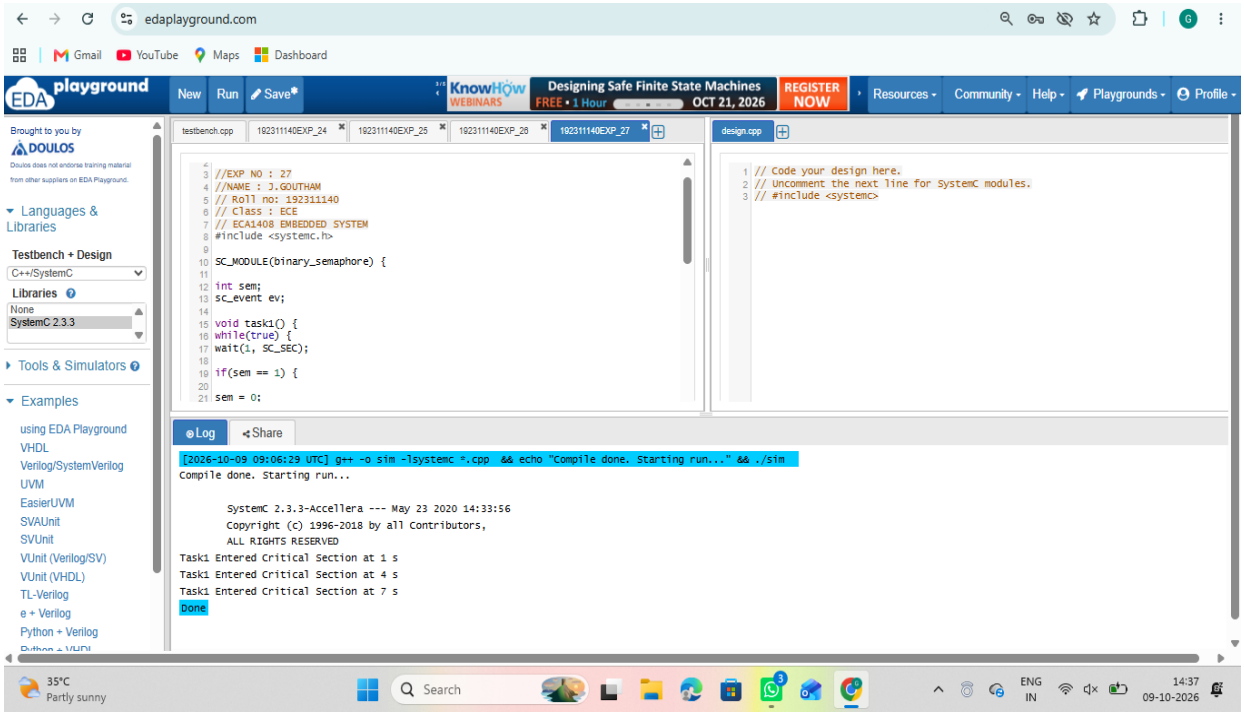
• SystemC

PROGRAM:

```
#include <systemc.h>
SC_MODULE(binary_semaphore) {
int sem; sc_event ev; void task1() {
while(true) { wait(1, SC_SEC); if(sem
== 1) { sem = 0;
cout << "Task1 Entered Critical Section at " << sc_time_stamp() << endl;
wait(2, SC_SEC); sem = 1;
ev.notify();
}
else { wait(ev);
}
}
}
```

```
void task2() {
while(true) {
wait(2, SC_SEC);
if(sem == 1) { sem
= 0;
cout << "Task2 Entered Critical Section at " << sc_time_stamp() << endl;
wait(1, SC_SEC); sem = 1;
ev.notify();
}
else { wait(ev);
}
}
}
SC_CTOR(binary_semaphore) { sem =
1;
SC_THREAD(task1); SC_THREAD(task2);
}
};
int sc_main(int argc, char* argv[]) {
binary_semaphore obj("Binary_Semaphore");
sc_start(10, SC_SEC); return 0;
}
```

OUTPUT:



RESULT:

Thus the Binary Semaphore was created using SystemC and verified.